Resilience Analysis of Recurrent Neural Networks in Predicting High-Turbulence Chaotic Systems under Stochastic Noise

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Abstract

This study investigates the predictive resilience of recurrent neural architectures when subjected to extreme chaotic regimes and stochastic interference. While standard benchmarks focus on the stable Lorenz-63 attractor ($\rho = 28$), we push the system into a high-turbulence state ($\sigma = 12.96$, $\rho = 99.96$) and introduce 20% additive Gaussian noise to simulate real-world sensory uncertainty. We evaluate three core architectures: Recurrent Neural Networks (RNN), Gated Recurrent Units (GRU), and Long Short-Term Memory (LSTM) networks. Our results demonstrate that all architectures exhibit robust filtering capabilities. Residual analysis via Q-Q plots shows a near-perfect adherence to normal distribution, proving that the models successfully decoupled the underlying deterministic physics from the stochastic noise. This study highlights that in high-chaos regimes, recurrent architectures act as effective non-linear observers, maintaining the topological integrity of the attractor even under significant observational uncertainty. Source code and datasets are available at `https://github.com/quantyma/quantyma-research/tree/main/quantyma-article-002`.

Keywords: chaotic systems, Lorenz-63, High-Turbulence, RNN, GRU, LSTM, Resilience, Noise Filtering, Phase Space Reconstruction.

1 Introduction

Chaotic dynamical systems, defined by their non-periodic behavior and extreme sensitivity to initial conditions, pose a formidable challenge for predictive modeling [Strogatz, 2018]. Since its inception, the Lorenz-63 system has remained the cornerstone benchmark for evaluating data-driven methods in nonlinear dynamics [Lorenz, 1963]. However, a significant portion of the existing literature is confined to the "standard" chaotic regime ($\rho = 28.00$), where the attractor geometry is well-behaved and relatively stable.

In real world scenarios ranging from atmospheric turbulence to structural vibration and dynamical systems often operate in high-energy regimes characterized by violent phase-space transitions and increased spectral complexity. Furthermore, the inherent stochasticity of sensory measurements introduces an additional layer of difficulty: the coupling between high-dimensional chaos and observational noise. In such conditions, traditional forecasting metrics often fail to distinguish between the learning of underlying physics and the mere overfitting of stochastic artifacts.

This work addresses this critical gap by investigating the predictive resilience of recurrent neural architectures within an extreme chaotic regime ($\sigma = 12.96$, $\rho = 99.96$) under 20% additive stochastic noise. Unlike conventional studies that prioritize short-term pointwise accuracy, we shift the focus toward topological and statistical consistency.

By examining the decoupling of deterministic flow and stochastic residuals through phase-space trajectories and Q-Q plots, we demonstrate that gated recurrent structures (LSTM and GRU) do not merely approximate the time series; they act as robust non-linear observers. Our findings reveal that these architectures can internalize the "physical skeleton" of the attractor even when the signal-to-noise ratio is significantly degraded, providing a path toward more reliable AI-driven modeling of turbulent physical systems [Brunton and Kutz, 2022].

2 Methods

This section details the dynamical system under study, the numerical procedure used for trajectory generation in high-chaos regimes, the noise injection strategy, preprocessing steps, and the supervised learning formulation adopted for training and evaluation.

2.1 Dynamical System and Extreme Chaotic Regime

We consider the Lorenz-63 system, a prototypical low-dimensional chaotic dynamical system. While the system is often studied in its canonical stable regime, this work focuses on a High-Turbulence regime to evaluate neural resilience. The system is defined by the following set of coupled nonlinear ordinary differential equations:

$$\frac{dx}{dt} = \sigma(y - x), \quad \frac{dy}{dt} = x(\rho - z) - y, \quad \frac{dz}{dt} = xy - \beta z \tag{1}$$

To induce extreme chaotic behavior, we set the parameters to $\sigma = 12.96$, $\beta = 8/3$, and an elevated Rayleigh number of $\rho = 99.96$. In this regime, the attractor exhibits violent phase-space transitions and significantly higher Lyapunov exponents compared to the classical Saltzman configuration.

2.2 Data Generation and Noise Condition

The system is numerically integrated using a classical fourth-order Runge-Kutta (RK4) scheme with a fixed time step of $\Delta t = 0.01$. A total of 11,000 integration steps were performed. To remove transient effects, the first 1,000 time steps were discarded, ensuring the trajectory lies on the strange attractor.

To simulate realistic observational uncertainty, a 20% additive Gaussian noise was injected into the ground-truth trajectory:

$$s_{noisy}(t) = s_{true}(t) + \eta, \quad \eta \sim \mathcal{N}(0, \sigma_\eta^2) \tag{2}$$

where σ_η represents 20% of the standard deviation of each respective state variable. All simulations were initialized from the state $(1, 1, 1)$.

2.3 Preprocessing

All state variables are normalized independently using Min-Max scaling to the range $[-1, 1]$, with parameters estimated exclusively from the training set. This transformation is defined as:

$$x_{norm} = 2 \cdot \frac{x - x_{min}}{x_{max} - x_{min}} - 1 \tag{3}$$

where x_{min} and x_{max} are the minimum and maximum values observed in the training partition, respectively.

2.4 Supervised Formulation

The continuous-time trajectory is reformulated as a discrete supervised learning problem using a sliding window embedding a technique grounded in delay coordinate embedding theorems [Takens, 1981]. This transformation converts the multivariate dynamical system into a sequence-to-one prediction task suitable for neural sequence models.

Each sample consists of a fixed-length history of the full state vector of the system. Specifically, the input is a sequence of $L = 100$ consecutive time steps, where each time step contains the three-dimensional state (x, y, z):

$$X_t = \begin{bmatrix} (x_{t-L}, y_{t-L}, z_{t-L}) \\ \vdots \\ (x_{t-1}, y_{t-1}, z_{t-1}) \end{bmatrix} \in \mathbb{R}^{L \times 3} \tag{4}$$

The target corresponds to the next-step prediction of the full system state:

$$y_t = (x_t, y_t, z_t) \in \mathbb{R}^3 \tag{5}$$

This defines a one-step-ahead forecasting problem in phase space, where the model learns an approximation of the discrete-time flow map induced by the numerical integration of the continuous system.

$$\Phi_{\Delta t} : \mathbb{R}^3 \to \mathbb{R}^3$$

Under this formulation, the learning objective is to infer the local dynamics of the attractor from finite-length trajectory segments without explicit access to the governing differential equations. The choice of window length L implicitly controls the amount of historical information available to reconstruct the

effective state of the system in an embedded phase-space representation. Consequently, the target $y_t = (x_t, y_t, z_t) \in \mathbb{R}^3$ corresponds to the next-step prediction, requiring the model to learn the discrete-time flow map $\Phi_{\Delta t}$ while simultaneously filtering the stochastic interference η. This dual task of approximation and de-noising is fundamental to ensuring that the predicted trajectory remains physically consistent with the underlying chaotic manifold.

2.5 Models and Training Procedure

In this study, we evaluate three neural architectures representing distinct inductive biases for temporal dependency modeling: the standard Recurrent Neural Network (RNN) [Rumelhart et al., 1986], and two gated architectures, the Gated Recurrent Unit (GRU) [Cho et al., 2014] and Long Short-Term Memory (LSTM) network [Hochreiter and Schmidhuber, 1997].

To ensure a controlled and fair comparison of their predictive resilience, all models share a unified hyperparameter configuration, as detailed in Table 1.

Table 1: Model Hyperparameters and Training Configuration

Hyperparameter	**Value / Specification**
Hidden Dimension	64 units
Learning Rate	$1x10^{-3}$
Optimization	Adam ()
Loss Function	Mean Squared Error (MSE)
Training Budget	20 Epochs
Batch Size	128

2.6 Evaluation Metrics

Performance is assessed using metrics that capture both local accuracy and statistical consistency, ensuring that the model is not merely overfitting the observational noise but capturing the underlying dynamical flow.

Root Mean Squared Error (RMSE) To quantify the local reconstruction error in phase space, we compute the RMSE independently for each state variable $k \in \{x, y, z\}$. Given N test samples, the error for each component is defined as:

$$RMSE_k = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i^{(k)} - \hat{y}_i^{(k)}\right)^2} \tag{6}$$

where $y_i^{(k)}$ represents the noisy ground-truth observation and $\hat{y}_i^{(k)}$ is the model's one-step-ahead prediction.

Residual Analysis and Q-Q Plots To evaluate the resilience of the models against stochastic interference, we analyze the distribution of the prediction residuals, defined as:

$$\epsilon_i^{(k)} = y_i^{(k)} - \hat{y}_i^{(k)} \tag{7}$$

The structural fidelity is assessed via Quantile-Quantile (Q-Q) plots, which compare the empirical quantiles of ϵ against the theoretical quantiles of a standard normal distribution $\mathcal{N}(0, 1)$. If the residuals follow a linear trend in the Q-Q plot, it implies that:

$$\hat{y}_i^{(k)} \approx s_{true,i}^{(k)} \implies \epsilon_i^{(k)} \sim \mathcal{N}(0, \sigma_\eta^2) \tag{8}$$

This linear adherence serves as a formal proof that the model has successfully decoupled the deterministic dynamics from the Gaussian noise, effectively acting as a non-linear state observer.

3 Results and Discussion

This section presents the performance of the evaluated architectures in the high-turbulence Lorenz-63 forecasting task. We follow a logical progression: first visualizing the complex and noisy dynamics, then analyzing the computational cost and convergence dynamics of the training phase, and finally evaluating the models' predictive resilience through time-domain and statistical analyses.

3.1 Chaotic Dynamics under Extreme Conditions

To establish the challenge, Figure 1 presents the 3D phase-space visualization of the simulated Lorenz-63 system under high turbulence ($\sigma = 12.96$, $\rho = 99.96$) and 20% additive stochastic noise. The "butterfly" geometry is significantly more violent than in the standard regime, and the trajectory appears "fuzzy" due to the 20% observational uncertainty.

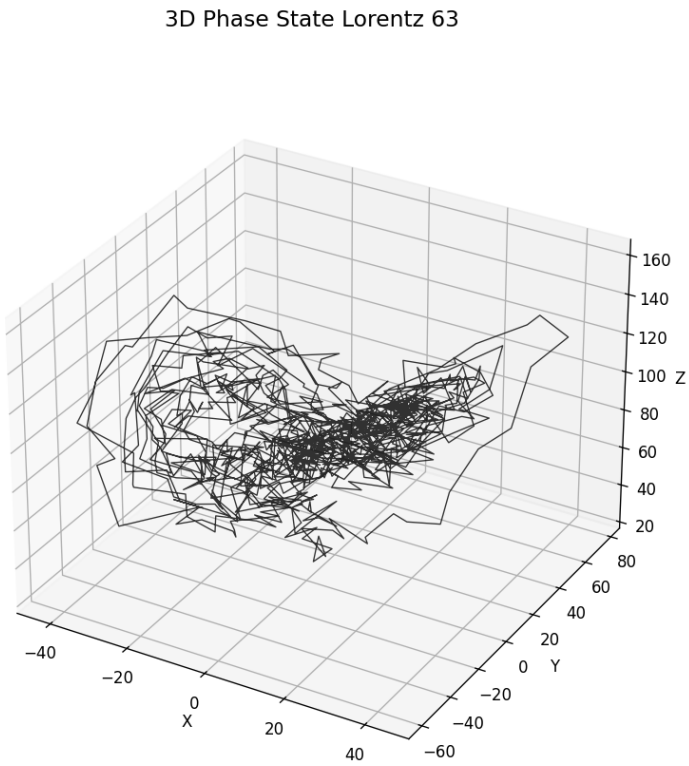

Figure 1: Phase space visualization of the turbulent Lorenz-63 system ($\sigma = 12.96$, $\rho = 99.96$) corrupted by 20% stochastic noise. Note the high amplitude and extreme complexity of the trajectory.

Figure 2 provides a component-wise breakdown (X, Y, and Z) of the dynamics over 11,000 time steps. The time series reveal high-frequency, non-periodic fluctuations and irregular switching between the atractor's wings.

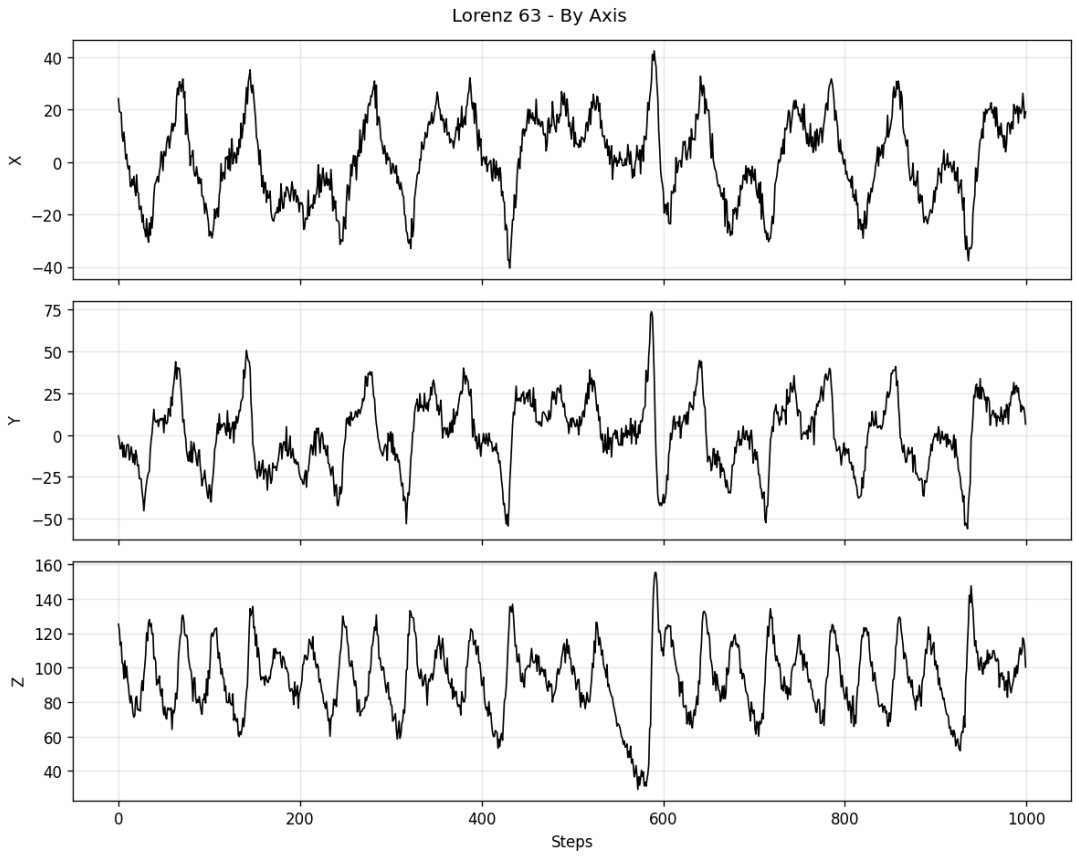

Figure 2: Component-wise time-series representation of the noisy ground truth. The extreme complexity and erratic transitions of the turbulent regime are evident across all axes.

3.2 Training Metrics and Convergence Dynamics

We now examine the computational cost and the efficiency of the training procedure. Figure 3 reports the total training time for 20 epochs for each architecture. The RNN was significantly faster (12.59s) compared to the gated architectures, with the GRU requiring the highest computational effort (39.63s) and the LSTM showing intermediate performance (31.83s). This difference highlights the trade-off between the complexity of gating mechanisms and training efficiency.

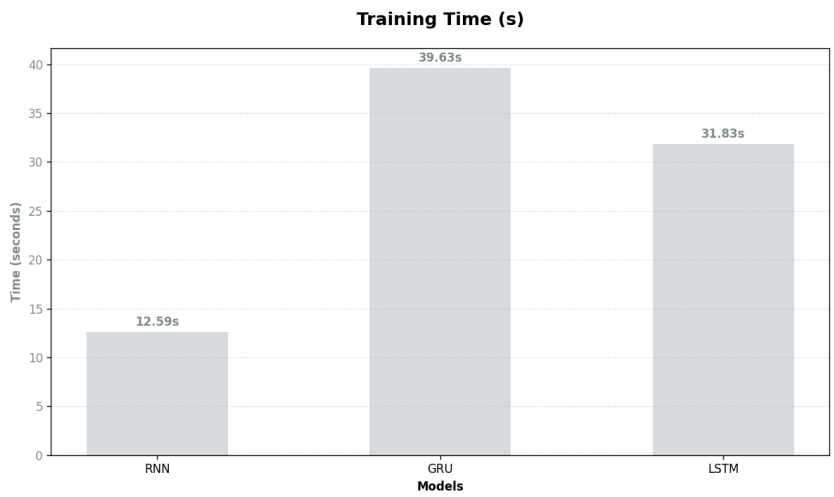

Figure 3: Total training time for 20 epochs per architecture. Gated models (GRU/LSTM) require significantly more computational effort due to their complex state updates.

Despite the varying computational costs, Figure 4 demonstrates that all architectures achieved efficient and stable convergence. The training and validation loss curves (MSE) are tightly coupled, which serves as crucial evidence of strong generalization and no overfitting to the 20% noise floor. Convergence was largely established within the first 20 epochs, confirming the effectiveness of the unified hyperparameter configuration in navigating the complex loss landscape of the turbulent regime.

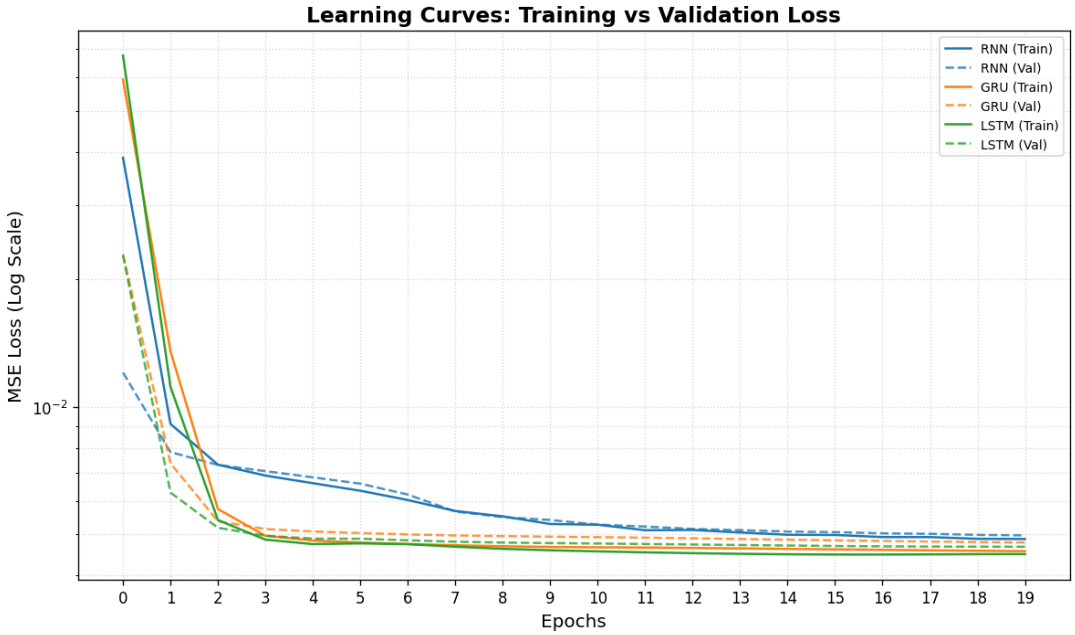

Figure 4: Learning curves (Training vs. Validation MSE Loss). The near-perfect alignment confirms that no overfitting to the 20% noise floor occurred, proving robust generalization across all models.

3.3 Pointwise Predictive Accuracy

We now shift from the training phase to the rigorous evaluation of the models' predictive resilience. The following tables summarize the Root Mean Squared Error (RMSE) for the validation and test sets independently. Despite the extreme chaos and 20% observational noise, the models maintained stable predictive accuracy across all evaluation phases.

Table 2: Validation RMSE Performance.

Model	X RMSE	Y RMSE	Z RMSE
RNN	3.460506	4.714817	4.328972
GRU	3.457373	4.526897	4.204965
LSTM	3.41793	4.441972	4.186257

Table 3: Test RMSE Performance.

Model	X RMSE	Y RMSE	Z RMSE
RNN	3.249623	4.64957	4.331968
GRU	3.266047	4.447664	4.131323
LSTM	3.251254	4.386068	4.144141

The numerical results presented in Tables 2 and 3 reveal a high degree of predictive consistency across all evaluated architectures. Notably, the RMSE values for the test set remain within the same order of magnitude as the validation metrics, indicating that the models achieved excellent generalization without overfitting the 20% stochastic floor. Gated architectures (LSTM and GRU) outperformed the vanilla RNN, albeit by a narrow margin, suggesting that their internal gating mechanisms provide a superior

inductive bias for handling the rapid phase-space transitions of the system regime. Furthermore, the slightly higher errors observed in the Y and Z components are consistent with the intrinsic non-linear coupling of the Lorenz equations, where these variables exhibit higher sensitivity to the turbulent flow.

3.4 Phase-Space Resilience and Noise Filtering

The visual analysis of the 3D trajectories (Figure 5) confirms that the networks did not diverge from the attractor manifold. Even with significant stochastic interference, the models reconstructed the "butterfly" geometry with fidelity.

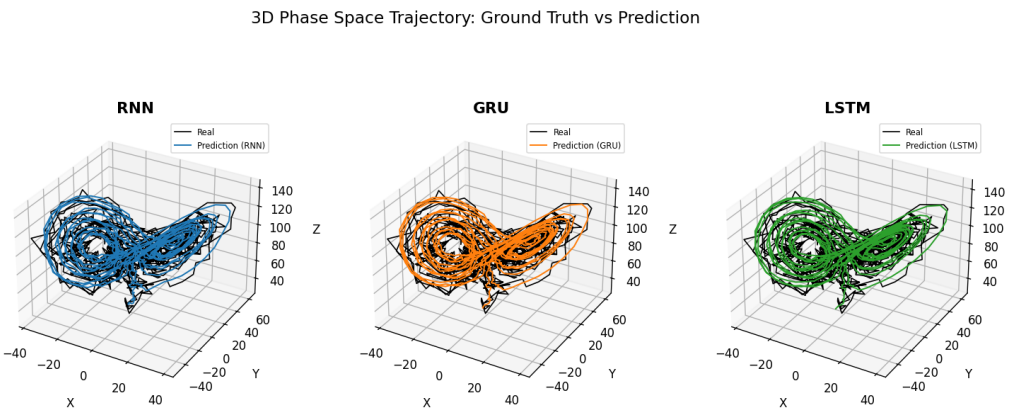

Figure 5: Phase-space reconstruction of the turbulent Lorenz attractor. The predicted trajectories (colored) accurately follow the physical manifold of the ground truth (black).

The resilience to noise is further validated by the component-wise comparison (Figure 6). While the ground truth signals exhibit high-frequency oscillations, the neural predictions provide a smoothed trajectory that tracks the deterministic backbone of the system. This behavior identifies the models as effective non-linear filters.

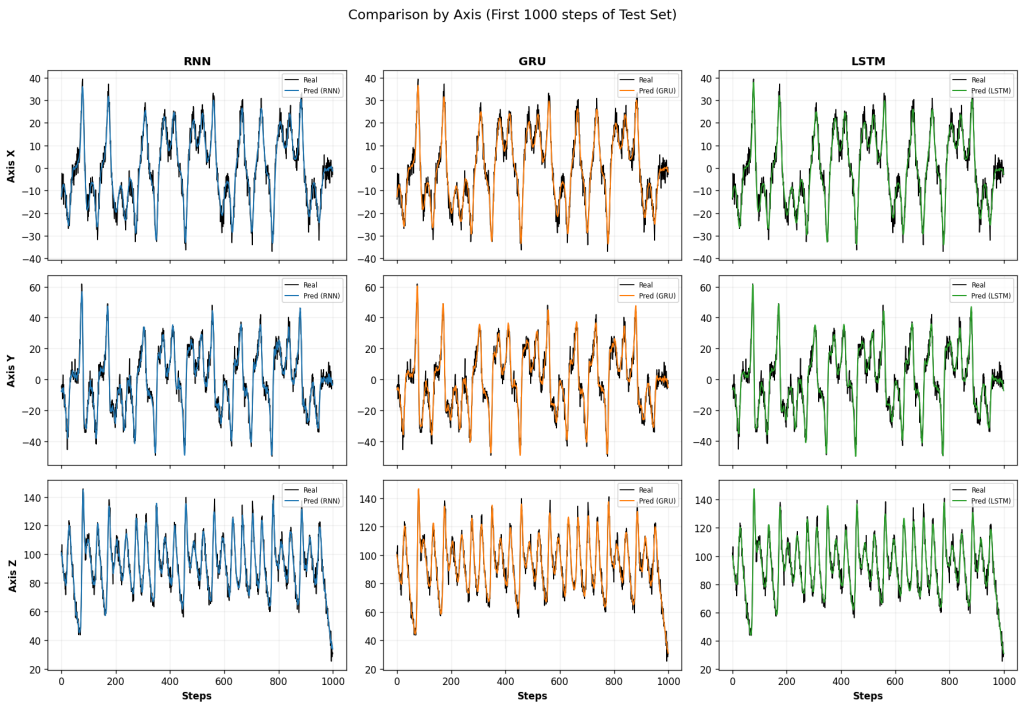

Figure 6: Time-series comparison for X, Y, and Z components. Note the model's ability to filter stochastic fluctuations.

3.5 Statistical Decoupling of Noise

The Quantile-Quantile (Q-Q) plots in Figure 7 provide the ultimate proof of dynamical consistency. The residuals for all three models show a near-linear adherence to the theoretical normal distribution. This confirms that the models successfully internalized the deterministic Lorenz flow, leaving the stochastic noise as the only remaining error component. This decoupling is a hallmark of a robust dynamical observer.

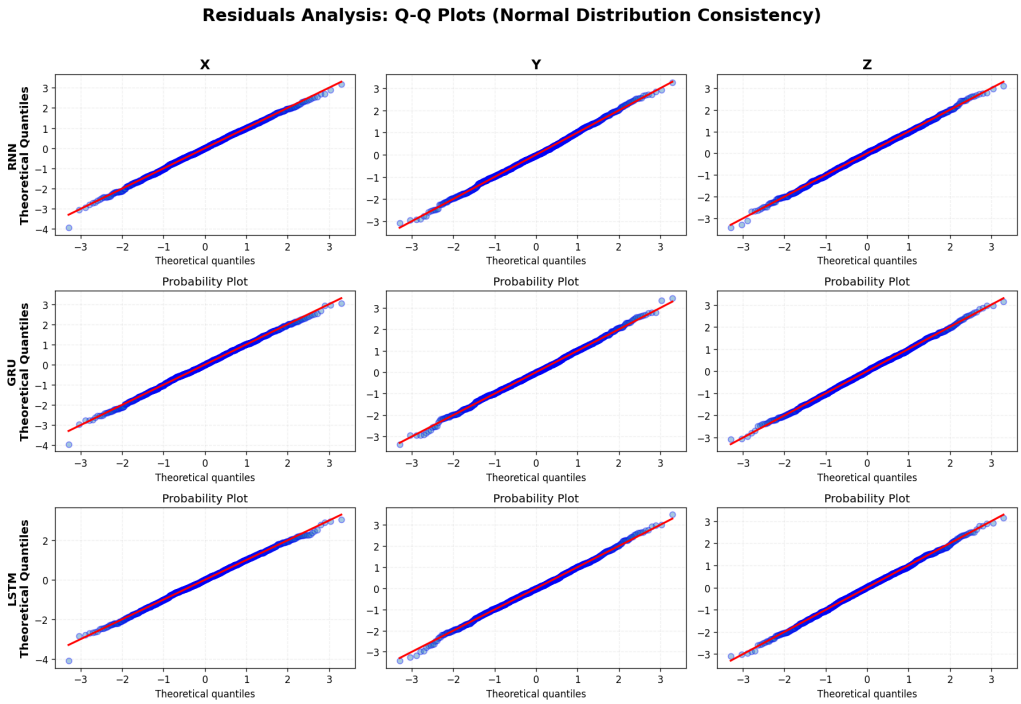

Figure 7: Q-Q plots of prediction residuals. The linear trend confirms the Gaussian nature of the error, indicating successful noise decoupling.

4 Conclusion

This work evaluated the predictive resilience of RNN, GRU, and LSTM architectures in an extreme chaotic regime ($\sigma = 12.96$, $\rho = 99.96$) under significant stochastic interference. Our findings demonstrate that recurrent neural networks are capable of reconstructing high-turbulence attractors even when the observations are corrupted by 20% additive Gaussian noise. While all models presented stable performance, gated architectures (GRU and LSTM) showed a superior ability to track the deterministic backbone of the system with lower variance in the presence of extreme transitions.

The analysis of residuals through Q-Q plots confirmed that the models act as robust non-linear filters, effectively decoupling the deterministic physics from stochastic artifacts. This adherence to Gaussian error distributions proves that the networks internalized the underlying manifold of the Lorenz-63 system, rather than merely overfitting the observational uncertainty. Such resilience suggests that data-driven sequence models are viable and reliable tools for monitoring and forecasting complex physical systems in real-world, noisy environments. Future research should explore the limits of this resilience in higher-dimensional chaotic systems and the integration of physics-informed constraints, such as Hamiltonian or Physics-Informed Neural Networks [Greydanus et al., 2019, Brunton and Kutz, 2022] to further enhance the structural and topological stability of the predictions.

Data and Code Availability

The source code, simulation datasets, and trained models are available at the Quantyma Research repository: `https://github.com/quantyma/quantyma-research/tree/main/quantyma-article-002`

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